

Cultural Analytics of Gender Representation in Film: A James Bond Case Study

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1. Introduction

Media has a long history of portraying gender in their stereotypical gender roles. Especially older blockbuster movies such as Indiana Jones [1] portray the man (Dr. Jones) as fearless “tough guy” who gets what he wants. Women on the other hand are portrayed as love interest, hysterical, and afraid. This portrayal seems out of date, and movies today, albeit still not perfect in their gender representation, would be unlikely to release with such messages. But when did this change occur? Was the change gradual, or did some events/movements bring forward bigger leaps of change?

In this document we investigate how the portrayal of Women has changed in the last decades. Specifically, we have chosen the James Bond movie series as subject of research. To make our findings quantifiable, the movies were analyzed using pretrained AI models, notably YOLO [2] and OpenAI CLIP [3]. Given the computer guided approach, we also investigate whether those models are suited for such analysis

In the process we have created an analysis framework, which can be used for other research questions. This resulting framework is corpus agnostic, and can be adapted to detect different personality traits. It is open source and free to use.

2. Dataset

The corpus chosen for our analysis is the James Bond 50th Anniversary DVD Collection. It was chosen for several reasons:

1. **Long Running Series** The Collection contains movies starting from 1962 to 2008 allowing for an observation period of 46 years
2. **Series Containing Many Movies** The Collection contains 22 movies. The release interval varies from movie to movie, but overall they were released mostly regularly. The longest time between releases was 6 years, but the average is around 2 years.

3. **Limiting Confounding Factors** By limiting our observation to one series, we limit confounding factors. James Bond movies all are of the same genre, mostly consist of the same slowly changing cast and crew, and follow similar plots. This makes comparison easier than comparing films of completely different genres (such as Indiana Jones, and Barbie)
4. **Notoriety For Gender Representation** The James Bond series notorious for it's representation of women, even coining the term Bond girl. The prior research on this Series also allows us to validate our methodology, so that it can be applied to other corpora

The movies were digitized using VLC. Due to copyright restrictions, they are not provided with the code-base. The code-base however is corpus agnostic and can be applied to any other collection of movies.

3. Methods

To retrieve the portrayal of women from the raw movie data several steps were required. First, the individual movies were split into their scenes and saved as images. Then, object detection was performed to get bounding boxes of persons in the scene images. Finally, trait and gender detection was performed.

3.1. Scene detection

First, the video data was split into scene data. This process was automated with PySceneDetect [4]. For every scene detected by PySceneDetect, one frame in the middle of the scene was saved as image (see Fig. 1). This resulted in 800-2000 frames per movie totalling 31000 frames.



Figure 1: scenes detected with PySceneDetect

3.2. Person detection

The next step applied person detection on every scene frame. This was performed with YOLO [2]. The pretrained yolo11 model was used, limited to only person detection. Furthermore the minimum confidence was set to 0.5. This helped to limit the detected persons to the people in the foreground.

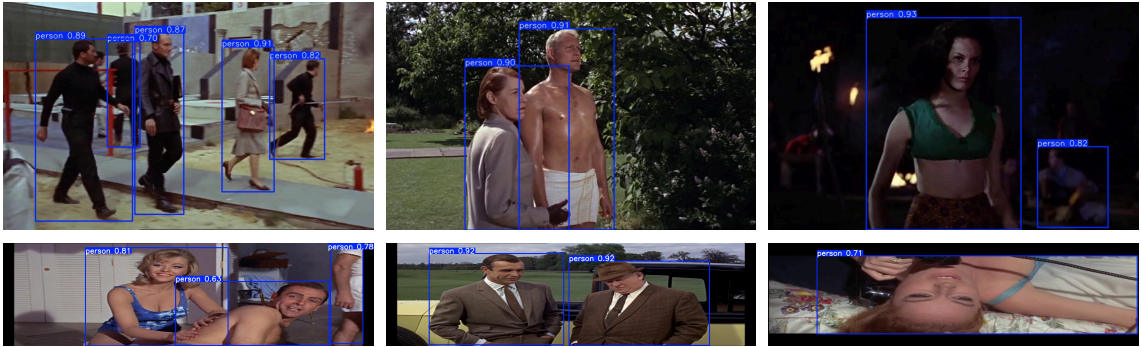


Figure 2: persons detected with YOLO11

3.3. Trait Detection

This step performs trait and gender detection for each bounding box, where YOLO detected a person. Trait and Gender detection are both performed using CLIP [3]. The images used are cropped along the bounding boxes to only contain the detected person. A batch size of 100 images was chosen for performance.

Text prompts were used to determine trait and gender (see Table 1). Multiple prompts were used per trait to lessen the effect of the individual prompts. To get a single prediction value from those, the mean of an ensemble was calculated, and the softmax was then calculated from both opposing means.

Table 1: prompt ensembles for trait detection

trait	negative	positive
gender	“a photo of a woman”	“a photo of a man”
agency	“a person overwhelmed with the situation”	“a person in control of the situation”
	“a person who is passive”	“a person who gets what they want”
	“a submissive person”	“a dominant person”
sexualization	“a sexualized person”	“a professional person”
	“a person in a swimsuit”	“a person in formal attire”
	“a scantily clad person”	“an adequately dressed person”

The following formula (see Fig. 3) used to transform logits to a score from 0 to 1, indicating how strongly the negative traits apply. To determine the degree of sexualization for instance, the mean would be calculated from the logits corresponding to “a sexualized person”, “a person in a swimsuit”, “a scantily clad person”, and the mean would be calculated from the logits corresponding to “a professional person”, “a person in formal attire”, “an adequately dressed person”. Then the softmax would be calculated from both means.

$$\text{softmax_trait} = \sigma\left(\overline{\text{logits}_{\text{trait}}^-}, \overline{\text{logits}_{\text{trait}}^+}\right) \quad (1)$$

Figure 3: scoring formula

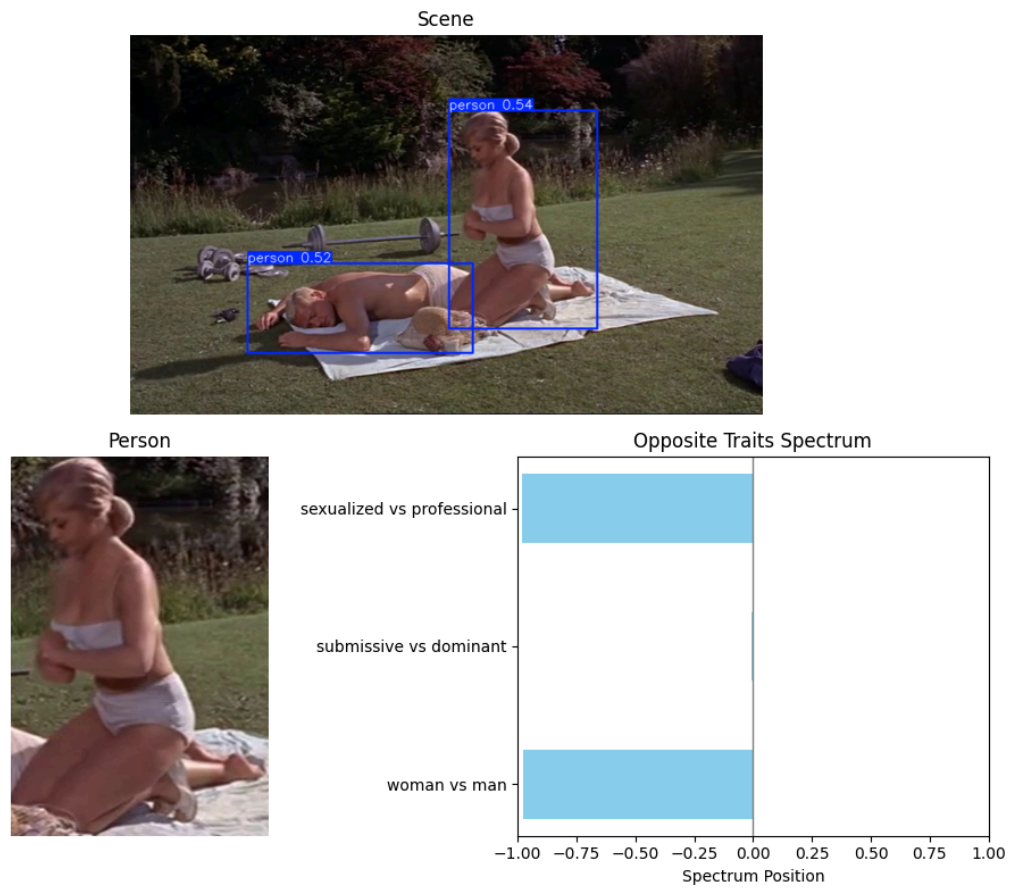


Figure 4: Two persons are recognized in this image. The woman is interpreted by the model as sexualized but neither dominant, nor submissive. In the movie, the woman is applying sunscreen for the man in the picture. The model however is missing the context, so it is missing

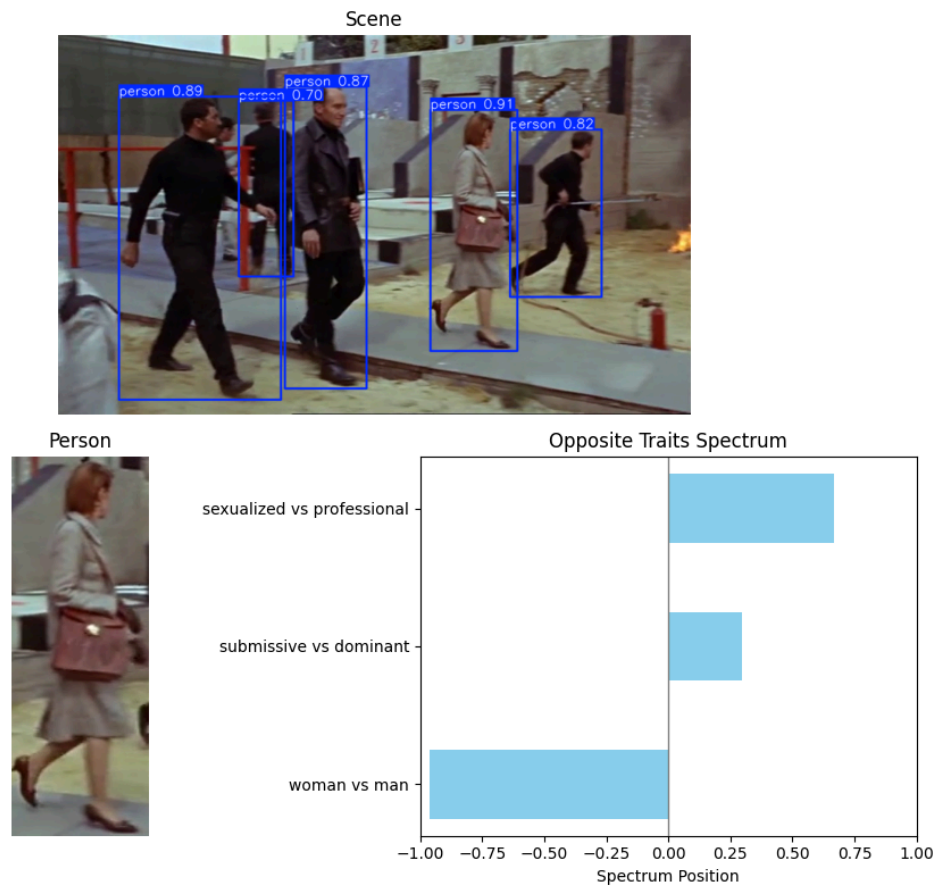


Figure 5: 5 Persons are recognize in this scene from *From Russia with Love*, one of them being the woman with the briefcase. She is interpreted by the model as a professional and dominant woman. The model is accurate, as the woman is the leader of a russian spy group currently visiting one of here training facilities.

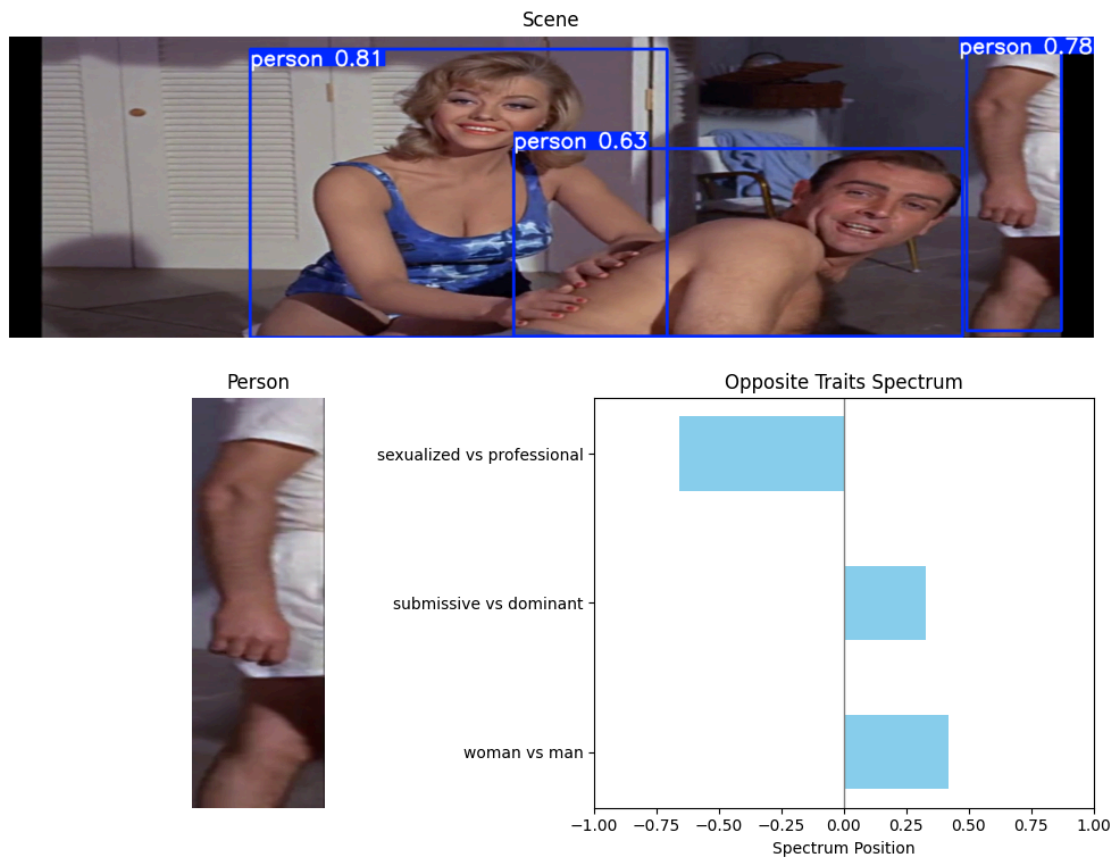


Figure 6: While the gender is correctly inferred, only the arm is visible. The model still predicts the degree of sexualization and dominance.

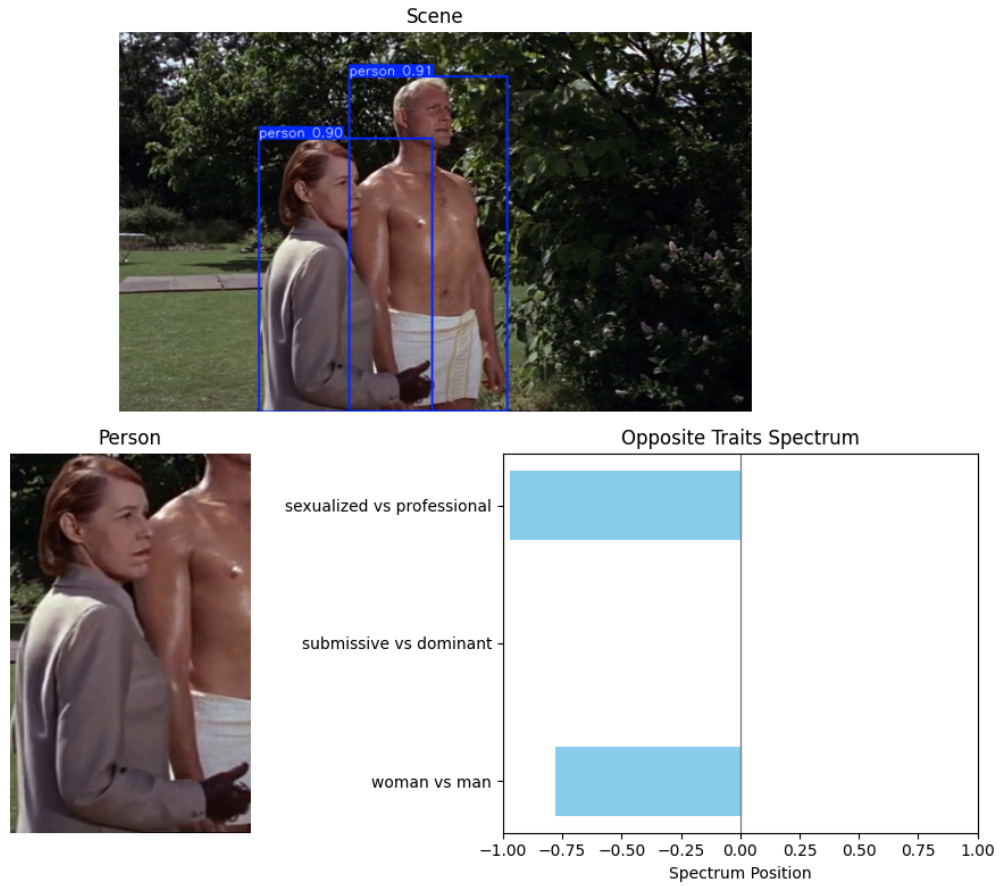


Figure 7: The gender is correctly inferred, but the degree of sexualization is clearly wrong. It's seems reasonable to assume that the naked man in the background has a high impact on the degree of sexualization, even though he is not the person of interest.

3.3.1. Limitations

This approach has limitations as can be seen in Fig. 7, where traits of a partial person are analyzed, and in Fig. 6, where a person in the background impacts the detected traits. Additionally, the model and the prompt bias can also contribute to a difference between genders. However, it should still be possible to monitor change over the period of the 22 films to observe change of representation within one gender.

4. Findings

In contrast to the representation of women in Indiana Jones, the models predicted that the women in James Bond were not significantly more helpless than men (see Fig. 8). This fits the subjective observation we have made, in which the women are commonly portrayed as confident and seductive. Regarding sexualized representation however, women were depicted significantly more sexualized than men (see Fig. 9). There is a trend toward less sexualization both in men and women. Where the sexualization scores for men and women in 1962 were at 0.26 and 0.56, in 2008 those scores dropped to 0.22 and 0.50.

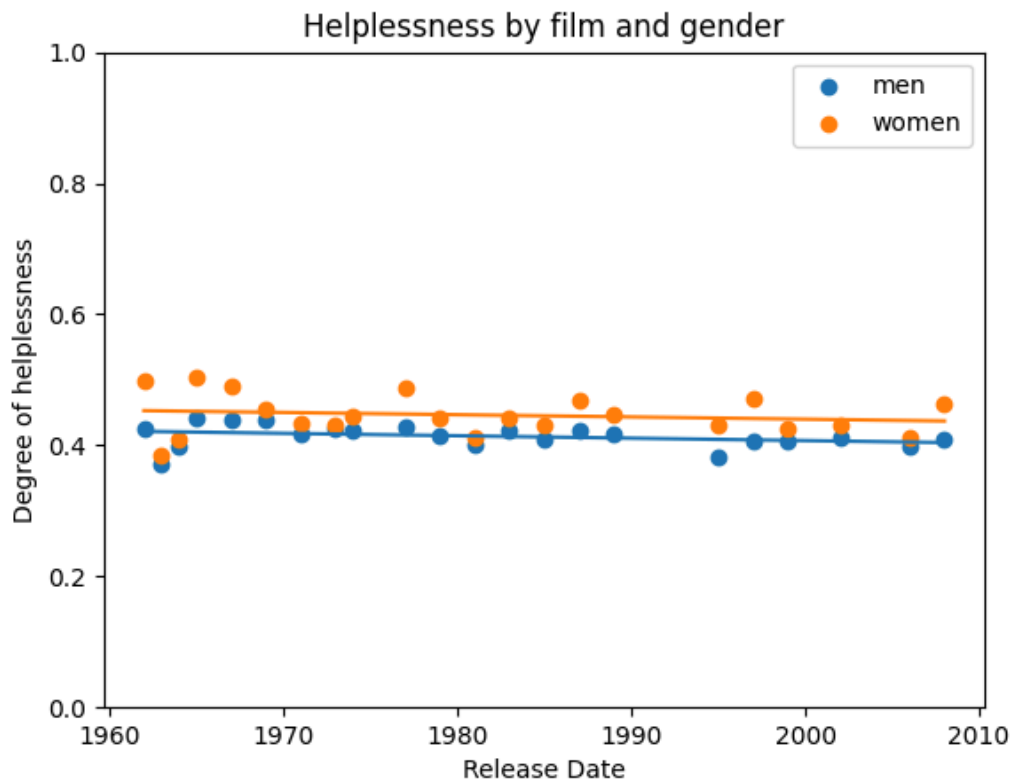


Figure 8: degree of helplessness in films by gender including trend lines. the

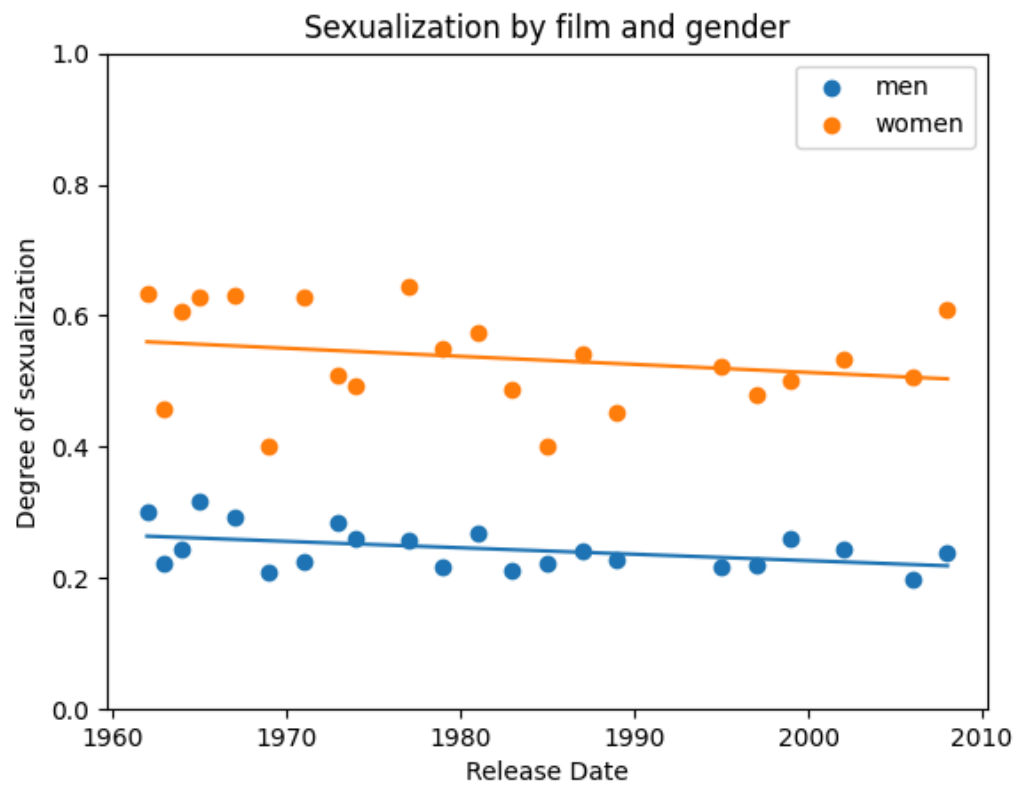


Figure 9: degree of sexualization in films by gender including trend lines

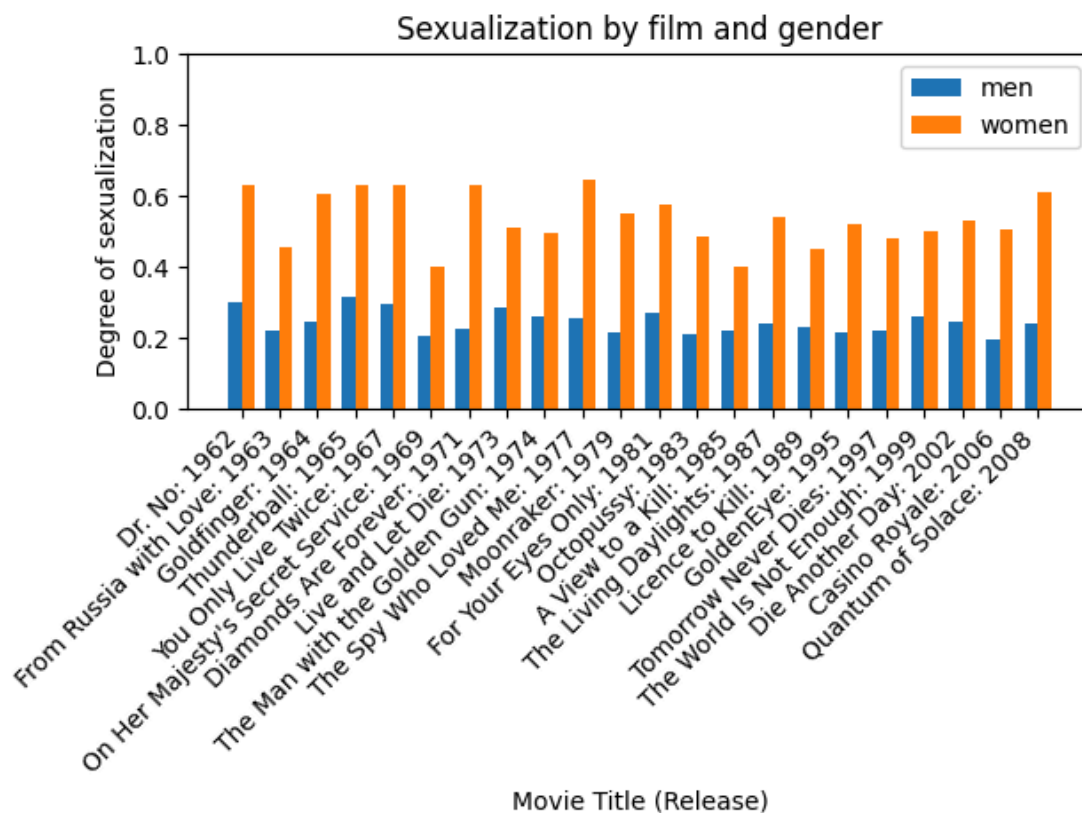


Figure 10: degree of sexualization by film title and gender

When plotting sexualization by titles instead, some outliers become apparent, notably *From Russia with Love: 1963*, *On Her Majesty's Secret Service: 1969*, and *A View to a Kill: 1987* (see Fig. 10). Those outliers can be well explained by the plot. In *From Russia with Love* James Bond is accompanied on his mission by a female spy whereas in other movies most women are only acquaintances he sleeps with. Thus there is more screentime to normalize the value. *On Her Majesty's Secret Service* also strongly differs from other James Bond movies, as the

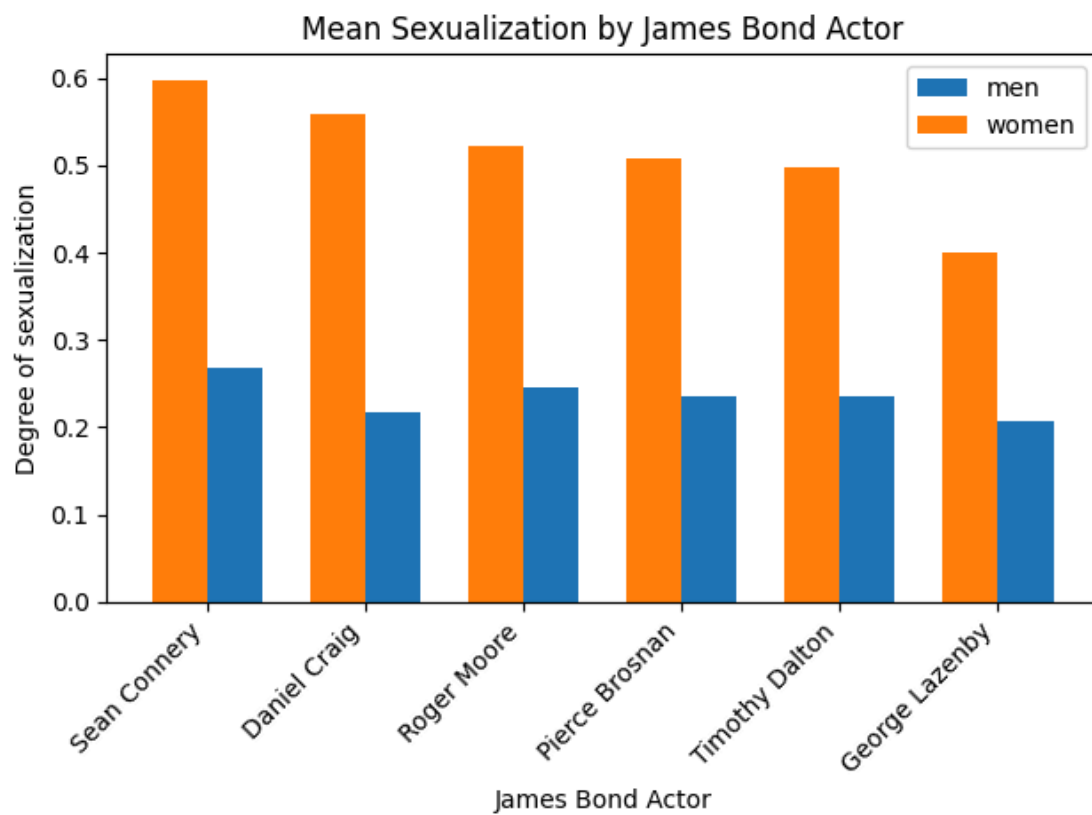


Figure 11: Movies with Sean Connery as James Bond actor have the most sexualized representation of women, whereas George Lazenby has the least sexualized representation.

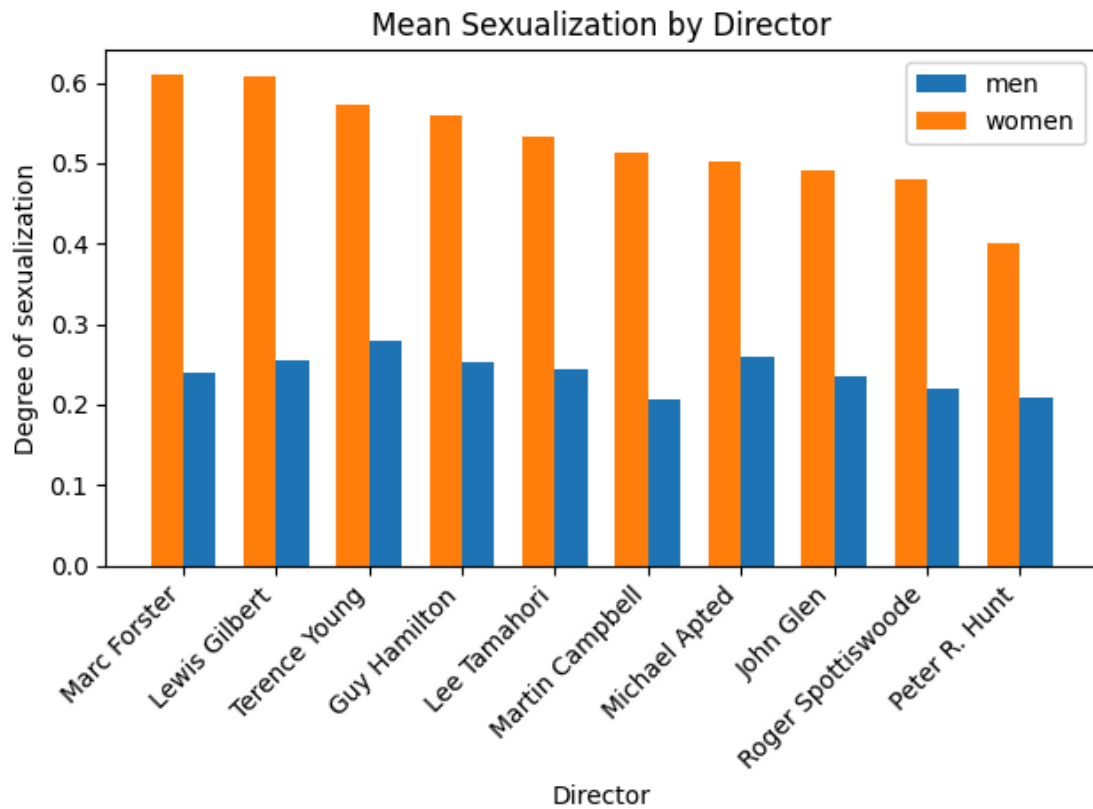


Figure 12: Marc Forster has the most sexualized depiction of women, whereas Peter R. Hunt the least sexualized depiction.

5. Further Work

Adding more traits, especially since there was not big difference between

Adding more Prompts, as some prompts may induce bias. A suit for instance is attributed to a man.

References

- [1] K. C. Schmidt, “Feminism and Indiana Jones: A Field Guide.” [Online]. Available: <http://dx.doi.org/10.1002/9781119740186.ch5>
- [2] “ultralytics — pypi.org.”

- [3] “GitHub - openai/CLIP: CLIP (Contrastive Language-Image Pretraining), Predict the most relevant text snippet given an image — github.com.”
- [4] B. Castellano, “Home - PySceneDetect — scenedetect.com.”